

Team Setup Guide: Google Colab + GitHub + Central AWS S3 Storage

Repository: <https://github.com/ShubhamJain17r/Amazon-ML-Hackathon.git>

Architecture Context:

In AWS, our team is **strictly using Amazon S3 for central cloud data storage**. We are **NOT** using SageMaker notebook instances (due to 0-instance quota restrictions and cost). Heavy computation (blocking, feature extraction, training) is executed on **Google Colab (Free GPU/High-RAM) or Local Machines**, connecting directly to our shared S3 bucket as a central database.

1. Quick Onboarding for Every Teammate (5 Minutes)

Step 1: Open Google Colab or Local JupyterLab

- If using **Google Colab**: Go to colab.research.google.com.
- Under **Runtime** $\rightarrow$ **Change runtime type**, select **Python 3** (choose **T4 GPU** or High-RAM if available).

Step 2: Configure AWS S3 Credentials in Colab Secrets

To avoid committing your AWS access keys into Git:

1. In Colab's left sidebar, click the **Key icon (Secrets)**.
2. Add the following secrets (toggle "Notebook access" ON):

- `AWS_ACCESS_KEY_ID`: [Provided by Shubham]
- `AWS_SECRET_ACCESS_KEY`: [Provided by Shubham]
- `AWS_DEFAULT_REGION`: `us-east-1`
- `S3_BUCKET_NAME`: [Provided by Shubham]
- `GH_TOKEN`: [Your personal GitHub Personal Access Token]

(If working on your local machine, simply export these in your shell or place them in a `.env` file).

2. Universal Colab Starter Cell

Copy and run this cell at the top of your notebook:

```

# 1. Install dependencies
!pip install -q boto3 s3fs pyarrow duckdb rapidfuzz lightgbm scikit-learn

import os
from google.colab import userdata

# 2. Inject AWS credentials into environment
os.environ["AWS_ACCESS_KEY_ID"] = userdata.get("AWS_ACCESS_KEY_ID")
os.environ["AWS_SECRET_ACCESS_KEY"] = userdata.get("AWS_SECRET_ACCESS_KEY")
os.environ["AWS_DEFAULT_REGION"] = userdata.get("AWS_DEFAULT_REGION", "us-east-1")
BUCKET = userdata.get("S3_BUCKET_NAME")

# 3. Clone repository and navigate inside
if not os.path.exists("Amazon-ML-Hackathon"):
    gh_token = userdata.get("GH_TOKEN")
    !git clone https://{gh_token}@github.com/ShubhamJain17r/Amazon-ML-Hackathon.git

%cd Amazon-ML-Hackathon

# 4. Verify S3 connection
import s3fs
fs = s3fs.S3FileSystem()
files = fs.ls(BUCKET)
print(f"✅ S3 Central Store Connected! Found folders: {files}")

```

3. How to Access Central S3 Data

Reading Filtered Data or Feature Sets

You can read directly from S3 without saving large files locally:

```

import pandas as pd

# Load features generated by Team A
train_features = pd.read_parquet(f"s3://{BUCKET}/features/sample_50k/train_candidates.parquet")
print(f"Loaded {len(train_features):,} rows directly from S3!")

```

Fast SQL Querying using DuckDB (Zero Memory Waste)

```
import duckdb

con = duckdb.connect()
con.execute(f"""
    INSTALL httpfs; LOAD httpfs;
    SET s3_region='us-east-1';
    SET s3_access_key_id='{os.environ["AWS_ACCESS_KEY_ID"]}';
    SET s3_secret_access_key='{os.environ["AWS_SECRET_ACCESS_KEY"]}';
""")

# Query parquet in S3 without loading into RAM
df = con.execute(f"SELECT * FROM 's3://{BUCKET}/features/sample_50k/train_candidates.parquet' WHERE label = 1 LIMIT 1000000")
```

4. Git Collaboration & Conflict Prevention

1. Always pull before starting work:

```
bash git pull origin main
```

2. Work strictly in your assigned notebook directory:

- Shubham: `notebooks/Shubham/`
- Karan: `notebooks/Karan/`
- Suhani: `notebooks/Suhani/`
- Vishal: `notebooks/Vishal/`

3. To commit and push:

```
bash git add notebooks/<YourName>/ git commit -m "feat: updated candidate generation logic" git push origin main
```

WARNING: **NEVER commit datasets, `.tsv`` files, `.parquet`` files, or `.pkl`` model weights to GitHub!** All data and model artifacts MUST be stored in the central S3 bucket.